

## ALBEMARLE OUTSON, United Kingdom

WMO: 032380

Lat: 55.0197N

Lon: 1.8817W

Elev: 146

StdP: 99.58

Time Zone: 0.00 (EUW)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|------------|-------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |            | 1%          |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-4.4 | (c)<br>-2.8 | (d)<br>-6.3                   | (e)<br>2.3 | (f)<br>-3.0 | (g)<br>-5.0 | (h)<br>2.5 | (i)<br>-1.2 | (j)<br>16.6           | (k)<br>7.9 | (l)<br>15.1 | (m)<br>8.2 | (n)<br>2.4            | (o)<br>290 | (p)<br>0.628 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.2                    | 22.8            | 17.2 | 21.1 | 16.4 | 19.6 | 15.4 | 18.1                | 21.6 | 17.1 | 20.0 | 16.2 | 18.7 | 3.5                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.7                            | 12.1 | 19.6 | 15.8 | 11.4 | 18.4 | 15.0 | 10.8 | 17.5 | 51.6          | 21.6 | 48.5 | 20.0 | 45.8 | 18.8 | 22.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|----------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |          |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%     | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 12.1          | (b) 10.3 | (c) 8.9 | (e) -6.6                   | (f) 26.0 | (g) 2.2            | (h) 2.1 | (i) -8.1                                           | (j) 27.6 | (k) -9.4   | (l) 28.8 | (m) -10.6  | (n) 30.0 | (o) -12.2  | (p) 31.6 |    |    |
| (a) 12.1          | (b) 10.3 | (c) 8.9 | (e) -6.9                   | (f) 20.0 | (g) 2.1            | (h) 1.2 | (i) -8.4                                           | (j) 20.8 | (k) -9.6   | (l) 21.5 | (m) -10.8  | (n) 22.2 | (o) -12.4  | (p) 23.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.7    | 3.6   | 3.8   | 5.1   | 7.4   | 9.9   | 12.8  | 14.8  | 14.5  | 12.5  | 9.5   | 5.9   | 3.9  |
|                                                                                  | DBStd       | 4.93   | 3.03  | 2.90  | 2.88  | 2.76  | 2.75  | 2.39  | 2.42  | 2.06  | 2.41  | 2.67  | 3.12  | 3.52 |
|                                                                                  | HDD10.0     | 1013   | 199   | 174   | 153   | 87    | 34    | 4     | 0     | 0     | 5     | 41    | 127   | 189  |
|                                                                                  | HDD18.3     | 3532   | 457   | 407   | 410   | 328   | 261   | 166   | 113   | 121   | 174   | 275   | 373   | 447  |
|                                                                                  | CDD10.0     | 528    | 0     | 0     | 2     | 8     | 32    | 88    | 149   | 139   | 81    | 24    | 3     | 1    |
|                                                                                  | CDD18.3     | 6      | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 1     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 37     | 0     | 0     | 0     | 0     | 1     | 4     | 26    | 4     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.8    | 4.8   | 4.1   | 4.4   | 3.9   | 3.8   | 3.2   | 3.1   | 3.3   | 3.6   | 3.5   | 4.1   | 4.2  |
| Precipitation                                                                    | PrecAvg     | 743    | 60    | 53    | 47    | 55    | 55    | 66    | 69    | 72    | 62    | 67    | 86    | 66   |
|                                                                                  | PrecMax     | 1045   | 155   | 113   | 102   | 139   | 90    | 168   | 143   | 170   | 158   | 135   | 160   | 162  |
|                                                                                  | PrecMin     | 531    | 9     | 10    | 11    | 10    | 17    | 16    | 11    | 13    | 19    | 16    | 27    | 31   |
|                                                                                  | PrecStd     | 118    | 33    | 28    | 22    | 41    | 22    | 42    | 33    | 34    | 37    | 30    | 38    | 31   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 11.9  | 12.7  | 15.0  | 18.9  | 22.1  | 23.7  | 26.1  | 23.3  | 22.6  | 17.4  | 14.7  | 12.7 |
|                                                                                  |             | MCWB   | 10.4  | 9.7   | 11.0  | 13.4  | 16.3  | 17.7  | 18.1  | 17.8  | 16.7  | 14.8  | 13.3  | 11.1 |
|                                                                                  | 2%          | DB     | 10.4  | 10.6  | 12.3  | 15.7  | 19.0  | 21.3  | 23.5  | 21.1  | 20.0  | 15.8  | 12.8  | 11.4 |
|                                                                                  |             | MCWB   | 9.0   | 8.8   | 9.5   | 11.3  | 14.4  | 16.1  | 17.7  | 16.8  | 15.8  | 13.6  | 11.4  | 10.0 |
|                                                                                  | 5%          | DB     | 9.3   | 9.4   | 10.9  | 13.8  | 16.6  | 19.2  | 21.4  | 19.6  | 18.1  | 14.7  | 11.5  | 10.4 |
|                                                                                  |             | MCWB   | 8.0   | 7.8   | 8.7   | 10.2  | 12.4  | 14.9  | 16.8  | 15.6  | 14.7  | 13.0  | 10.2  | 9.1  |
|                                                                                  | 10%         | DB     | 8.2   | 8.2   | 9.7   | 12.4  | 14.9  | 17.7  | 19.6  | 18.4  | 16.8  | 13.6  | 10.4  | 9.3  |
|                                                                                  |             | MCWB   | 7.0   | 6.8   | 7.6   | 9.2   | 11.7  | 14.1  | 15.7  | 14.9  | 13.8  | 11.9  | 9.2   | 8.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.6  | 10.4  | 11.5  | 13.7  | 17.0  | 19.0  | 19.8  | 18.8  | 17.9  | 15.4  | 13.3  | 11.2 |
|                                                                                  |             | MCDB   | 11.7  | 12.0  | 14.5  | 17.8  | 21.8  | 22.8  | 24.4  | 22.2  | 21.4  | 16.7  | 14.4  | 12.4 |
|                                                                                  | 2%          | WB     | 9.2   | 9.1   | 10.1  | 12.1  | 14.7  | 16.7  | 18.3  | 17.5  | 16.3  | 14.2  | 11.7  | 10.3 |
|                                                                                  |             | MCDB   | 10.2  | 10.6  | 11.9  | 15.1  | 17.9  | 19.5  | 22.2  | 20.2  | 19.0  | 15.3  | 12.5  | 11.3 |
|                                                                                  | 5%          | WB     | 8.2   | 8.1   | 8.9   | 10.7  | 13.3  | 15.5  | 17.2  | 16.6  | 15.2  | 13.2  | 10.5  | 9.3  |
|                                                                                  |             | MCDB   | 9.2   | 9.3   | 10.6  | 13.3  | 16.0  | 18.5  | 20.6  | 18.6  | 17.2  | 14.5  | 11.3  | 10.3 |
|                                                                                  | 10%         | WB     | 7.1   | 7.0   | 7.9   | 9.5   | 12.0  | 14.5  | 16.3  | 15.7  | 14.3  | 12.2  | 9.4   | 8.2  |
|                                                                                  |             | MCDB   | 8.2   | 8.1   | 9.6   | 11.9  | 14.3  | 17.2  | 19.1  | 17.6  | 16.2  | 13.4  | 10.3  | 9.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.5   | 5.3   | 6.6   | 7.7   | 7.9   | 8.0   | 8.2   | 7.6   | 7.5   | 6.2   | 5.3   | 5.1  |
|                                                                                  |             | MCDBR  | 5.8   | 6.6   | 8.2   | 10.4  | 10.9  | 10.9  | 11.0  | 9.0   | 9.5   | 7.1   | 5.9   | 6.1  |
|                                                                                  |             | MCWBR  | 5.3   | 5.2   | 5.7   | 6.5   | 6.5   | 6.5   | 6.0   | 5.2   | 5.7   | 5.4   | 5.5   | 5.6  |
|                                                                                  | 5% WB       | MCDBR  | 5.7   | 6.3   | 7.5   | 9.6   | 10.0  | 9.9   | 10.0  | 7.9   | 8.3   | 6.5   | 5.8   | 6.1  |
|                                                                                  |             | MCWBR  | 5.4   | 5.2   | 5.7   | 6.4   | 6.4   | 6.3   | 5.9   | 5.1   | 5.6   | 5.4   | 5.5   | 5.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.314  | 0.330 | 0.361 | 0.384 | 0.383 | 0.383 | 0.384 | 0.374 | 0.355 | 0.338 | 0.313 | 0.297 |      |
|                                                                                  | taud        | 2.301  | 2.375 | 2.307 | 2.267 | 2.316 | 2.339 | 2.342 | 2.384 | 2.443 | 2.446 | 2.363 | 2.270 |      |
|                                                                                  | Ebn at Noon | 612    | 750   | 805   | 837   | 860   | 862   | 853   | 840   | 809   | 732   | 613   | 540   |      |
|                                                                                  | Edh at Noon | 56     | 74    | 98    | 117   | 118   | 117   | 115   | 103   | 86    | 68    | 52    | 46    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.56   | 1.20  | 2.32  | 3.76  | 4.88  | 4.86  | 4.67  | 3.84  | 2.74  | 1.49  | 0.71  | 0.42  |      |
|                                                                                  | RadStd      | 0.07   | 0.13  | 0.27  | 0.37  | 0.30  | 0.42  | 0.42  | 0.27  | 0.23  | 0.14  | 0.07  | 0.04  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (50 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page